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Correct Answer

Next Question

Practice Exam - 1

Question 14 of 117



A 34-year-old man comes to clinic for cascade genetic screening, as his father had been recently diagnosed with a Lamin A/C associated cardiomyopathy. The patient is asymptomatic and feels generally well. He is active, jogging 30 minutes five times a week. He takes no medications. His blood pressure is 114/68 mmHg, heart rate is 78 beats per minute. Examination is unremarkable.

Electrocardiography demonstrates normal sinus rhythm, and first-degree AV block, with a PR interval of 280 ms. Echocardiography demonstrates a mildly dilated left ventricle, and an ejection fraction of 42%. The patient's genetic testing indicates the same mutation in LMNA (the gene encoding Lamin A/C) as his father.

Which of the following represents the next best step?

- A. Initiate metoprolol succinate 25 mg daily
-  B. Continue surveillance of left ventricular ejection fraction with repeat echocardiography in one year
- C. Institute home blood pressure monitoring
-  D. Recommend placement of implantable cardioverter defibrillator

Commentary

References

Testing Point: The Board-certified AHFTC specialist should know that genetic cardiomyopathy associated with Lamin A/C mutations portend a high risk of cardiac sudden death, and ICD can be considered at ejection fractions up to 45% in patients with other risk factors.

Answer: D. Recommend placement of implantable cardioverter-defibrillator.

Lamin A/C mutations portend a very high risk of cardiac sudden death and represent an important cause of dilated cardiomyopathy with prominent conduction system disease. The disease typically manifests with cardiac dysfunction or skeletal myopathy. Inheritance is autosomal dominant. Clinical features include heart block, LV dysfunction, and arrhythmia. Arrhythmic risk is high. The 2017

AHA/ACC/HRS guidelines recommend that patients with Lamin A/C mutation who have 2 of the 4 following features: NSVT, LVEF < 45%, nonmissense mutation, and male sex consider placement of an ICD. Since this patient meets two of the four criteria, ICD placement should be considered.

Answers A-C: Continued surveillance of LV dysfunction, home blood pressure monitoring, and initiation of metoprolol succinate are all reasonable additional steps. However, these steps do not offer sufficient protection against cardiac sudden death, a critical component of the natural history of lamin cardiomyopathy.

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